1. Item 7: Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

(Operate on numerical, graphical, statistical, and other mathematical representations)

- Highly relevant: The core task is to compare four decimal numbers (1.79, 1.796, 1.82, 1.783), which constitutes direct manipulation of numerical data.
- Students must understand place value in decimals (e.g., recognizing that 1.79 is equivalent to 1.790) and perform accurate comparisons—a demonstration of proficiency with the decimal number system and basic numerical handling.
- While no arithmetic computation is required, this precise interpretation and comparison of decimals exemplifies the "manipulation of numbers" emphasized in Item 7.

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2. Item 3: Implement a given strategy to determine a mathematical solution

(Carry out a specified or clearly implied procedure to reach a solution)

- Highly relevant: The problem implies a clear, straightforward strategy: "Compare all engine capacities and identify the smallest." Students simply need to execute this procedure.
- Although the strategy isn't explicitly stated, it is obvious within the multiple-choice context, fitting the description of implementing a given (or readily apparent) approach.

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3. Item 1: Perform a simple calculation

(Carry out basic computational tasks)

- Moderately relevant: Strictly speaking, the task does not require arithmetic operations (e.g., addition or multiplication). However, comparing decimals is often categorized broadly as a "simple calculation" in educational assessment contexts.
- It involves foundational numerical judgment rather than symbolic computation, but still aligns loosely with the intent of Item 1.

4. Item 10: Make sense of, relate and use a variety of representations when interacting with a problem

(Interpret and utilize different forms of representation)

- Moderately relevant: The data are presented in a tabular format, and students must extract information from a specific column ("engine capacity") and interpret it as comparable numerical values.
- This requires understanding and using "tables" as a standard data representation, which falls under the scope of Item 10.

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5. Item 11: Switch between different representations in the process of finding solutions

(Translate among representations during problem solving)

- Weakly relevant: A student might mentally rewrite "1.79" as "1.790" to align decimal places for comparison with "1.783." This internal adjustment constitutes a minor shift in numerical representation.
- However, this conversion is very basic and does not involve switching between fundamentally different representational forms (e.g., graph to equation, table to verbal description), so relevance is limited.

1. 第7条：Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

（操作数字、图形、统计信息等）

- 高度相关：本题核心任务是比较四个小数（1.79, 1.796, 1.82, 1.783），这属于对数值数据的直接操作。
- 学生需理解小数位数的意义（例如，1.79 实际等于 1.790），并进行精确比较，这体现了对十进制数系统的掌握和数值处理能力。
- 此过程不涉及复杂计算，但要求准确解读和比较数值——正是第7条所强调的“manipulate numbers”。

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2. 第3条：Implement a given strategy to determine a mathematical solution

（执行给定策略以得出数学解）

- 高度相关：题目隐含了一个清晰的解题策略：“比较所有发动机排量，找出最小值”。学生只需按此策略执行即可。
- 虽然策略未明说，但在选择题语境下，该策略是标准化且直接的，符合“implement a given (or obvious) strategy”的描述。

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3. 第1条：Perform a simple calculation

（进行简单计算）

- 中等相关：严格来说，本题不需要算术运算（如加减乘除），但“比较小数”可视为一种基础数值判断活动，常被广义归类为“simple calculation”。
- 在教育测评语境中，小数比较常与简单计算归为一类基础技能，因此此项具有一定相关性。

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4. 第10条：Make sense of, relate and use a variety of representations when interacting with a problem

（理解、关联并使用多种表征方式）

- 中等相关：数据以表格形式呈现，学生需从结构化表格中提取特定列（engine capacity）的信息，并将其转化为可比较的数值。
- 这涉及对“表格”这一数学/数据表征的理解和使用，属于第10条的范畴。

5. 第11条：Switch between different representations in the process of finding solutions

（在解题过程中切换不同表征）

- 弱相关：学生可能在心中将“1.79”转换为“1.790”以便与“1.783”对齐比较，这涉及小数表示的内部调整，可视为一种表征转换（从简写到标准位数）。
- 但这种转换非常基础，且未涉及图表、公式等多元表征间的切换，因此相关性较弱。